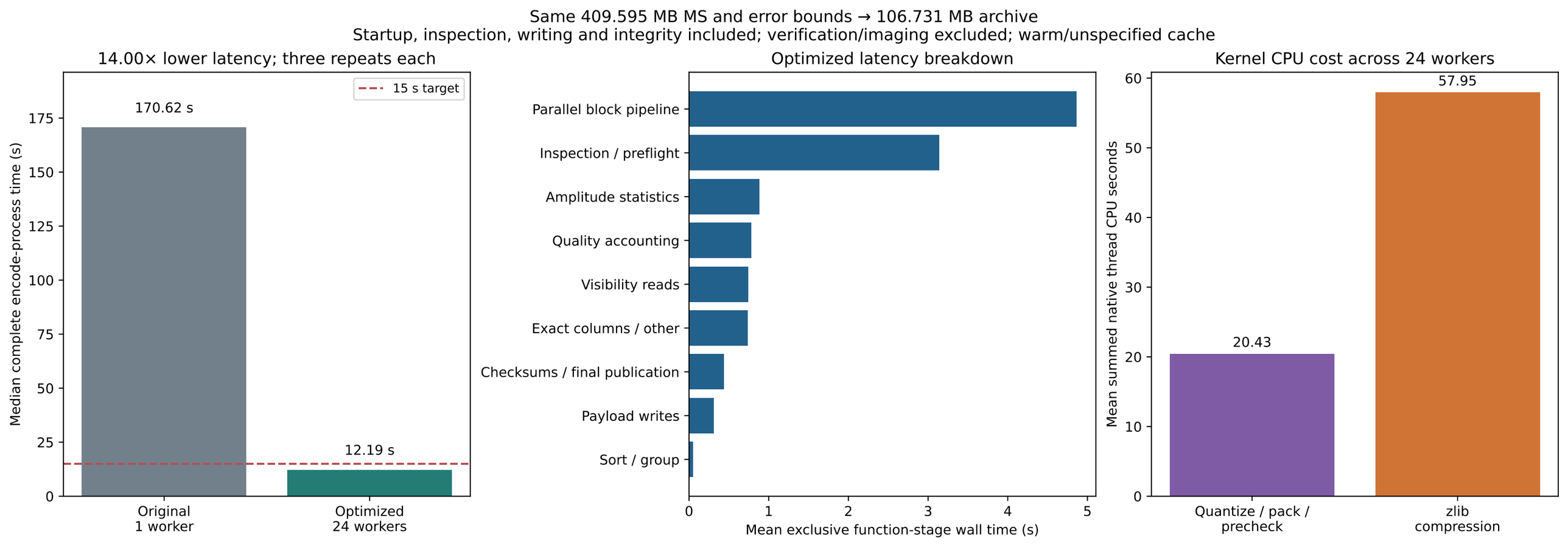
Same 409.595 MB MS and error bounds → 106.731 MB archive
Startup, inspection, writing and integrity included; verification/imaging excluded; warm/unspecified cache
14.00× lower latency; three repeats each
Median complete encode-process time (s)
15 s target
170.62 s
12.19 s
Original
1 worker
Optimized
24 workers
Optimized latency breakdown
Parallel block pipeline
Inspection / preflight
Amplitude statistics
Quality accounting
Visibility reads
Exact columns / other
Checksums / final publication
Payload writes
Sort / group
Mean exclusive function-stage wall time (s)
Kernel CPU cost across 24 workers
Mean summed native thread CPU seconds
20.43
57.95
Quantize / pack /
precheck
zlib
compression